CSE 445 · MACHINE LEARNING · SECTION 8

RT-IoT2022 Multi-Class IoT Attack Detection

An End-to-End Machine Learning Pipeline for Network Intrusion Detection across 12 Distinct IoT Traffic & Cyberattack Categories

FACULTY & TA

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Presentation Roadmap

Overview

01 · AGENDA

01 · Data & Variables

RT-IoT2022 dataset structure, 123,117 flows, 85 raw columns, and taxonomy of statistical flow features.

02 · Cleaning & Prep

Deduplication, dropping zero-variance constants, label encoding, and strictly isolated 70/15/15 train-val-test split.

03 · Model 1: LogReg

Multinomial Logistic Regression baseline, softmax probabilities, cross-entropy loss, and L2 regularization.

04 · Model 2: MLP

Deep feed-forward neural network architecture, ReLU activations, Adam optimizer, and backpropagation.

05 · Model 3: Random Forest

Ensemble bootstrap aggregation, decision tree voting, Gini impurity splits, and feature subspace sampling.

06 · Results & Selection

Comparative validation/test evaluation, Macro F1-score (0.96), confusion analysis, and deployment limitations.

Problem Context & Project Objectives

IoT Vulnerabilities in Modern Networks

Internet of Things (IoT) endpoints—such as smart sensors, industrial controllers, and surveillance systems—possess constrained computational power, making them prime targets for botnet recruitment, volumetric denial-of-service, and credential attacks.

- **Real-Time Threat Classification:** Distinguishing benign IoT operations from sophisticated multi-vector attacks in sub-second timeframes.
- **12-Class Granularity:** Moving beyond simple binary flags to pinpoint specific attack behaviors (e.g., SYN Floods, PortScans, BruteForce).

RESEARCH GOAL: Build a scalable, non-leaking ML pipeline with high precision across both majority and rare minority attack signatures.

Target Attack Taxonomy (12 Classes)

The RT-IoT2022 dataset provides a realistic representation of modern malicious and legitimate IoT network behaviors:

`Benign / Normal` `DOS_SYN_Hping` `Thing_Discover` `ARP_Poisoning`
`Nmap_PortScan` `Nmap_OSScan` `Nmap_UDPScan` `MQTT_Publish`
`Metasploit_Brute` `Nmap_FinScan` `Slowloris_DoS` `Hydra_SSH`

Class Imbalance Implication: High-volume DoS flows vastly outnumber stealthy reconnaissance scans, requiring **Macro-averaged F1-scores** rather than raw accuracy alone.

Dataset & Feature Variables Architecture

Feature Groups & Statistical Descriptors

Each recorded row represents a single unidirectional/bidirectional network conversation flow summarized into 84 statistical parameters:

Feature Group	Representative Variables Used	Behavioral Signal
Packet Volume	`fwd_pkts_tot`, `bwd_pkts_tot`, `data_pkts_tot`	Payload vs empty overhead
Rate & Speed	`fwd_pkts_per_sec`, `flow_pkts_per_sec`	Volumetric flooding rates
Header Stats	`fwd_header_size_min/max/tot`	Abnormal protocol headers
TCP Control Flags	SYN, FIN, RST, ACK, PSH flag counts	Half-open scans & teardowns

Key Discriminating Variables

Empirical analysis shows that IoT cyberattacks create distinct topological fingerprints across specific feature subsets:

- **SYN & RST Flag Ratios:** Extreme spikes in forward SYN with zero ACK bytes characterize SYN flood attacks (e.g., `DOS_SYN_Hping`).
- **Inter-Arrival Times (IAT):** Microsecond variance in `fwd_iat_min` vs `fwd_iat_max` exposes automated scripted port sweeps.
- **Asymmetric Byte Flow:** Zero backward byte counts (`bwd_data_pkts_tot = 0`) identify unacknowledged scanning probes.

Data Cleaning & Preprocessing Pipeline

STEP 1

Deduplication & Hygiene

- Removed identical duplicate rows to prevent artificial accuracy inflation.
- Dropped constant, zero-variance columns carrying zero variance signal (e.g., `bwd_URG_flag_count`).

STEP 2

Encoding & Scaling

- Encoded multi-class target labels into numeric integers [0–11].
- Standard scaling applied for gradient-based models (Logistic Regression & MLP) to normalize wide packet-count variances.

STEP 3

Leakage-Free 70/15/15 Split

Training (70%): Fit models & parameter estimation

Validation (15%): Model comparison & hyperparameter tuning

Test (15%): One-time final unbiased generalization evaluation

Model 1: Multinomial Logistic Regression

05 · LINEAR BASELINE

Mathematical Working Principle

Acts as the linear benchmark. For 12 classes, it models the posterior probability of class k using the Softmax link function over linear combinations of input features x:

$$P(y = k \mid x) = \frac{e^{w_k^T x + b_k}}{\sum_{j=1}^{12} e^{w_j^T x + b_j}}$$

Optimized by minimizing the multi-class Cross-Entropy loss with an L_2 weight penalty.

Parameters & Hyperparameters Used

Parameter	Setting / Value	Functional Impact
`multi_class`	`'multinomial'`	Joint probability estimation over 12 classes
`solver`	`'lbfgs' / 'saga'`	Quasi-Newton gradient optimization
`C (Inverse Reg.)`	`1.0`	Controls L2 regularization penalty
`max_iter`	`1000`	Ensures convergence on scaled features

Strengths & Role: Highly interpretable feature coefficients; achieved **99.05% validation accuracy**, confirming strong linear separability in flow features.

Model 2: Multilayer Perceptron (MLP)

Feed-Forward Architecture & Activation

A multi-layer feed-forward artificial neural network trained via backpropagation to model complex non-linear boundary manifolds:

$$h^{(l)} = \mathrm{ReLU} W^{(l)} h^{(l-1)} + b^{(l)}$$

- **Input Layer:** 84 input neurons corresponding to cleaned numerical flow variables.
- **Hidden Layers:** Fully connected dense layers with ReLU activation to avoid vanishing gradients.
- **Output Layer:** 12 neurons paired with Softmax activation.

Training Parameters & Optimization

Hyperparameter	Setting	Purpose
`hidden_layer_sizes`	`(100, 50)`	Hierarchical feature extraction depth
`activation`	`'relu'`	$f(z) = \max(0, z)$ non-linearity
`optimizer`	`'adam'`	Adaptive moment estimation gradient descent
`learning_rate_init`	`0.001`	Initial step size across weight updates
`early_stopping`	`True`	Monitors validation loss to prevent overfitting

Performance: Achieved **99.60% validation accuracy**, effectively learning subtle non-linear attack interactions.

Model 3: Random Forest Classifier (Ensemble)

Ensemble Architecture & Bagging

An ensemble of decorrelated decision trees built on bootstrap data samples with random feature sub-selection at each split node:

$$I_G(t) = 1 - \sum_{k=1}^{12} p_k^2, \quad \text{Prediction:}\hat{y} = \text{mode}h_1(x), \ldots, h_B(x)$$

- **Bootstrap Aggregation:** Decreases model variance without increasing bias.
- **Feature Subspace Sampling:** $\sqrt{p} \approx 9$ random features per split prevent dominant feature lock-in.

Hyperparameters & Key Attributes

Parameter	Configuration	Role in Stability
`n_estimators`	100 trees	Stabilizes ensemble majority voting
`criterion`	`'gini'`	Fast computation of split impurity
`max_features`	`'sqrt'`	Maximizes tree decorrelation
`n_jobs`	`-1`	Parallelized multi-core training

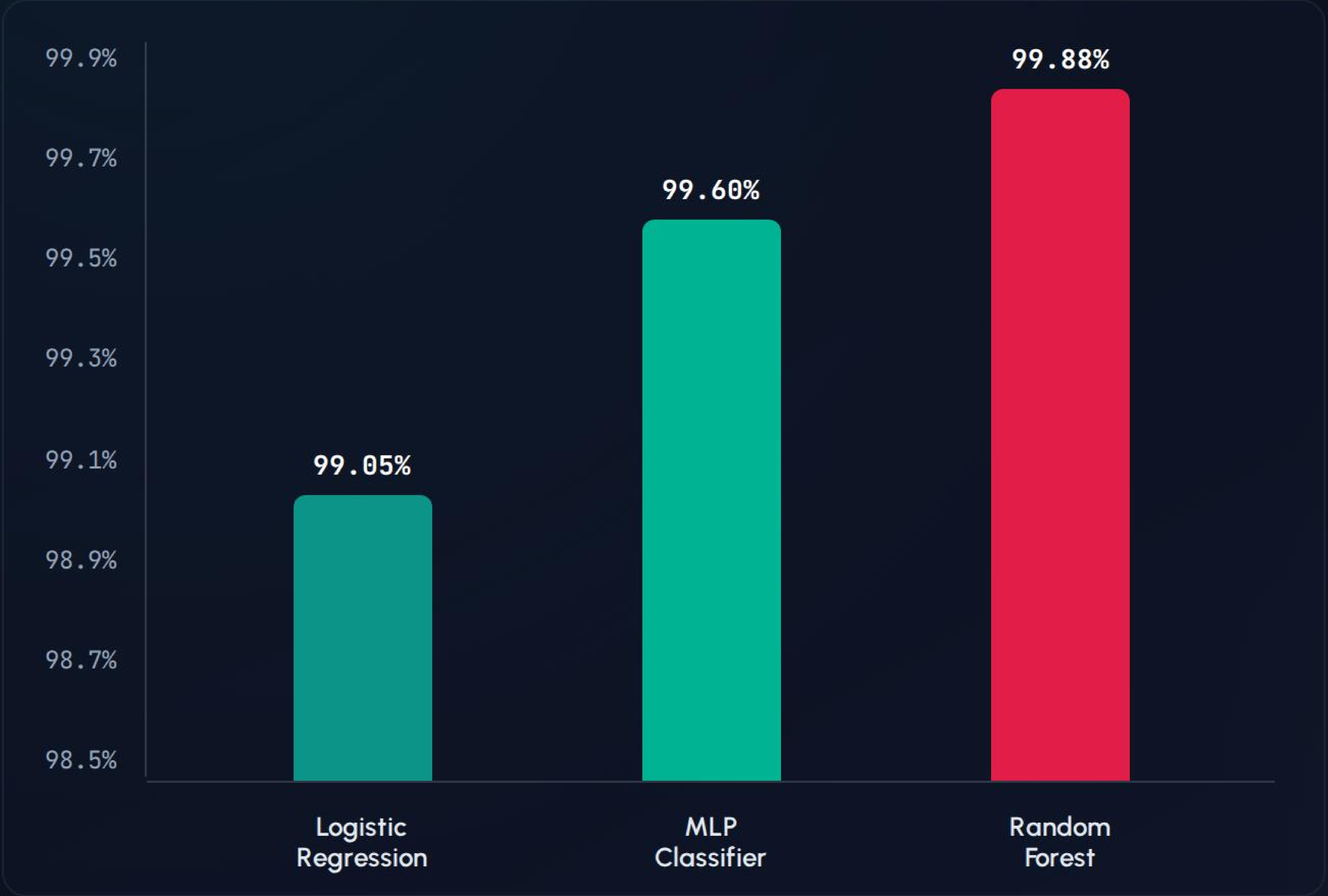
Why RF Excels in IoT: Invariant to monotonic scaling, robust to heavy outliers in packet bursts, and resilient to feature collinearity.

Comparative Model Specification Matrix

Model Dimension	Logistic Regression	MLP Classifier	Random Forest (Best)
Learning Paradigm	Linear probabilistic classifier	Feed-forward deep neural network	Ensemble Bagging of Decision Trees
Decision Boundary	Linear hyperplanes in R^{84}	Non-linear smooth manifold	Orthogonal axis-aligned partitions
Key Hyperparameters	`C=1.0, solver='lbfgs'`	`hidden=(100,50), lr=0.001`	`n_estimators=100, max_feat='sqrt'`
Scaling Dependency	Strictly required (StandardScaler)	Strictly required (StandardScaler)	Invariant (Tree thresholding)
Interpretability	High (direct weight coefficients)	Low (black-box latent weights)	Moderate (MDI feature importance)
Validation Accuracy	99.05%	99.60%	**99.88% (Selected)**

05 · RESULTS

Validation Accuracy: Model Comparison



Random Forest wins

All three models scored above 99% — the dataset's flow-level features are highly discriminative. Random Forest was selected for final testing.

99.88%

Best validation accuracy

Final Test Evaluation — Random Forest

99.83%	**0.96**	**Random Forest**
Final Test Accuracy	Macro F1-score	Selected Model

A Macro F1-score close to overall accuracy shows the model generalizes across all 12 classes — not just the majority class — including minority attack types.

Class-Wise Performance & Confusions

Strong Across Categories

- **High Precision & Recall:** Maintained >0.95 across nearly all 12 traffic categories, including small minority attack classes.
- **Imbalance Resilience:** Preprocessing, robust stratified splitting, and ensemble feature bagging prevented the majority class bias.

Where Confusion Happens

- **Similar Traffic Signatures:** Rare misclassifications occur primarily between attack variants sharing similar flow footprints (e.g., Nmap OSScan vs PortScan).
- **Confusion Matrix:** Validated exact true-vs-predicted pairwise intersections in the project notebook.

Critical Analysis & Real-World Limitations

Dataset & Class Constraints

- **Lab Separability:** High accuracy (99%+) reflects clean, synthetic flow environments; production network jitter and noise will reduce live accuracy.
- **Rare Class Sensitivity:** Minority classes with very few samples can still exhibit instability in recall metrics under distribution shifts.
- **Zero-Day Vulnerability:** The static supervised model cannot classify novel attack vectors outside its 12 predefined classes.

Operational & System Constraints

- **Explainability Tradeoff:** Random Forest ensembles lack the straightforward parameter interpretability required for audited security compliance.
- **IoT Edge Resource Footprint:** Storing and evaluating 100 deep decision trees in real-time introduces memory and latency overhead on low-power IoT microcontrollers.
- **Tuning Scope:** Models utilized standard optimized validation parameters; automated Bayesian hyperparameter search remains open for future work.

Conclusion & Future Research Directions

Random Forest: Recommended Model

- **Robust Supervised Pipeline:** Successfully cleaned, engineered, and evaluated 123,117 network flows across 12 attack categories with zero data leakage.
- **Champion Performance:** Achieved **99.83% Test Accuracy** and **0.96 Macro F1-Score** on the held-out test set.

Future Roadmap

- **Model Pruning & Quantization:** Compressing Random Forest and MLP models for low-latency edge deployment on ARM/ESP32 IoT devices.
- **Adversarial & Zero-Day Defense:** Integrating unsupervised anomaly detectors (e.g., Isolation Forests, Autoencoders) to capture unseen zero-day attacks.

Thank you! Questions & Discussion